Medical Surgical - Lecture

UPHSD - LP | LECTURE BY: Sir Rainier Del Rosario

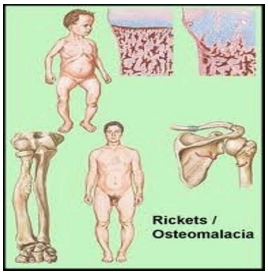
Lesson 3

Metabolic Bone Disorders

- Osteomalacia
- Osteoporosis
- Paget's disease of the bone

1 OSTEOMALACIA

- a metabolic bone disease results from:
- Deficiency of activated vitamin D (calcitriol)↓
  - ↓ *failed Calcium absorption from GI tract*
  - ↓ *Calcium in the bone or mineralization*
  - = causing soft and weak bone



SIGNS AND SYMPTOMS

- Pain
- tenderness to touch
- bowing of the bones, and pathologic fractures.
- On physical examination, skeletal deformities (spinal kyphosis and bowed legs)
- Waddling or limping gait.
- Muscle weakness, and unsteadiness

CAUSES

- failed calcium absorption (↓ activated vit. D deficiency or malnutrition)
- Body utilizes available calcium to combat acidosis caused by kidney disease = results to excessive loss of calcium
- Gastrointestinal disorders (eg, celiac disease, chronic biliary tract obstruction, chronic pancreatitis, small bowel resection)- fats are inadequately absorbed, loss of vit. D along with fat soluble vitamins and being excreted in the feces with fatty acids
- liver diseases fail to convert vit. D to active form
- Hyperparathyroidism d/t hypocalcemia parathyroid gland secretes↑PTH = targets bone to break down & ↑calcium in the blood / ↑excretion of phosphate via urine
- Prolonged use of antiseizure medication (eg, phenytoin, phenobarbital) can cause DRUG-INDUCED osteomalacia as adverse effect

CLINICAL ASSESSMENT

X-ray:
- *(+) generalized demineralization of bone*
- *(+) compression fracture on vertebrae*

Laboratory studies:
- ↓ ***serum calcium*** *= ↑ moderately elevated alkaline phosphatase*
- *(calcium binds with phosphate and vice versa)*

Urinalysis: ↓ calcium and creatinine

Bone biopsy: (+) increased amount of osteoid (unmineralized bone tissue that forms prior to the maturation of bone tissue)

MEDICAL MANAGEMENT

- increased doses of vitamin D (50,000 IU once or twice a week for 6-12 months then 1,000 IU per day), along with supplemental calcium
- Exposure to sunlight for ultraviolet radiation to transform a cholesterol substance (7-dehydrocholesterol) present in the skin into vitamin D
- Diet with adequate protein and increased calcium and vitamin D.
- Instruct about dietary sources of calcium and vitamin D (eg, fortified milk and cereals, eggs, chicken livers).

DIETARY HABITS FOR BONE HEALTH to protect against skeletal demineralization

- TAKE CALCIUM SUPPLEMENT WITH VITAMIN D, take with meals or beverage high in VITAMIN C to promote absorption. Recommended dose should be split and not taken as a single dose.
  - *Side effects: constipation, abdominal distention*
- **VITAMIN D** – is needed for calcium absorption. Adult: 400-1000 IU daily
  - *SOURCES of Vit. D: egg yolk, saltwater fish and liver. Fortified Vit. D Milk and cereals*
- **CALCIUM** – men, age: above 50-70 years old: 1000 mg of daily and above 70: 1200 mg daily, women, age: above 50 years old 1200 mg daily. Adverse effect of high dose of calcium is Renal Calculi
  - *SOURCES of calcium: low fat milk, yogurt, cheese. Steamed broccoli, canned salmon with bones. Fortified calcium in orange juices, cereals, bread*
- **Plus**: regular weight-bearing exercises promotes bone formation. *E.g. 20-30 minute aerobics, bone- stretching exercises daily*
- *Avoid: alcohol and tobacco*

Medical Management

- Monitor serum calcium levels

- Assist patient to change positions and handle the patient gently
- Reduce discomfort and pain
- Pillows to support the body
- Some persistent orthopedic deformities may need to be treated with braces or surgery (eg, osteotomy may be performed to correct long bone deformity).

Treat the underlying cause to resolve and correct the disorder:

- (+) kidney disease prevents activation of absorbed VIT. D give activated form of VIT. D supplement (CALCITRIOL)
- (+) malabsorption increase dose of VIT. D intake, give supplemental calcium, exposure to sunlight because UV radiation transforms cholesterol substance in the skin into VIT. D

2 PAGET DISEASE "osteitis deformans"

- Cause is UNKNOWN
- is a disorder of localized rapid bone turnover, most commonly affecting the skull, femur, tibia, pelvic bones, and vertebrae.
- 2% to 3% of the population older than 50 years old
- Slightly greater in aging men than in women
- Family history is noted
- is a chronic disease in which the bones gradually become thick and also soft, causing them to bend.

BONE becomes: enlarged, deformed, weak

- Localized PROLIFERATION OF OSTEOCLASTS causing rapid **excessive bone resorption** (breakdown of old bone) skull, femur, tibia, pelvic bones, and vertebrae followed by increased OSTEOBLASTIC ACTIVITY (**excessive bone formation**) = (+) bone enlargement BUT structurally WEAK bone turnover process continues creating a disorganized pattern of bone development = (+) DEFORMITY

Clinical Manifestations – initially no symptoms

- Skull involvement: Cranium is enlarged, face small and triangular appearance
- Impaired hearing – cranial nerve compression
- Sclerotic changes and cortical thickening of the long bones
- Bowing of femurs and tibia
- Waddling gait
- Spine is bent forward with chin rests on the chest and shortened trunk, thorax becomes immobile during respiration
- Bone warmth and tenderness d/t increased bone vascularity
- Pain is mild to moderate, deep and aching
- High cardiac output due to highly vascular regions in the bone

Assessment and Diagnostic Findings

- Increased osteoblastic activity:
  - *URINALYSIS: ↑ hydroxyproline*
  - *BLOOD: ↑ (ALP) alkaline phosphatase*
- normal blood calcium levels.
- X-rays: (+) local demineralization and bone overgrowth in mosaic pattern (jigsaw pattern)
- Bone scans and Bone biopsy demonstrate the extent of the disease and aid in the differential diagnosis.

Medical Management

- Administration of NSAIDs
- Gait problems from bowing of the legs are managed with **walking aids, shoe lifts, and physical therapy.**
- Weight control to reduce stress on weakened bones and misaligned joints.
- Diet adequate in calcium and vitamin D and periodic monitoring.

PHARMACOLOGIC THERAPY

1. **Calcitonin**, a polypeptide hormone
   a. retards bone resorption by decreasing the number and availability of osteoclasts
2. **Bisphosphonates**, such as etidronate disodium (Didronel) and alendronate sodium (Fosamax)
   a. Cornerstone of Paget therapy, produce rapid reduction in bone turnover and relief of pain. Reduces serum ALP and urine hydoxyproline level
3. **Plicamycin (Mithracin)**, a cytotoxic antibiotic given via IV
   a. pain reduction and decreases serum calcium, alkaline phosphatase, and urinary hydroxyproline levels. To monitor: kidney liver and bone marrow functions during therapy

COMPLICATIONS OF PAGET'S DISEASE:

1. **FRACTURE** – immobilization and reduction
2. **ARTHRITIS**- severe degenerative arthritis require total joint replacement
3. **HEARING LOSS** – hearing aids and communication techniques

3 Osteoporosis

- A disease of the bone characterized by a decrease in the bone mass and density with a change in bone structure
- *Complication: Fracture*

Pathophysiology

- Normal homeostatic bone turnover is altered rate of bone **RESORPTION is greater than bone FORMATION** reduction in total bone mass reduction in bone mineral density prone to **FRACTURE**

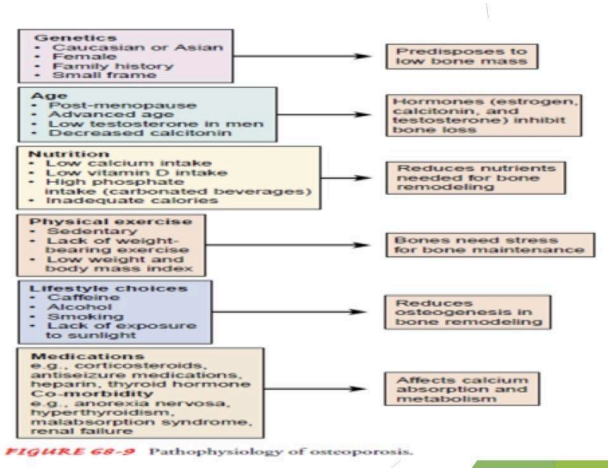


FIGURE 68-9 Pathophysiology of osteoporosis.

Individual Risk Factors

- **GENETICS** - Female, Caucasian, non- Hispanic or Asian
- **AGE** - Increased age
- **NUTRITION** - Low weight and body mass index
- Estrogen deficiency or menopause
- Family history
- Low initial bone mass
- Contributing, coexisting medical conditions (eg, celiac disease) and medications (eg, corticosteroids, antiseizure medications)

Lifestyle Risk Factors

- Diets low in calcium and vitamin D
- Cigarette smoking
- Use of alcohol and/or caffeine
- Lack of weight-bearing exercise
- Lack of exposure to sunshine

Risk-Lowering Strategies

- Increased dietary calcium and vitamin D intake
- Smoking cessation
- Alcohol and caffeine consumption in moderation
- Regular weight-bearing exercise regimen
- Walk or exercise out of doors

TYPES

1. **Primary Osteoporosis**- advanced age, post-menopausal in women and men later in life. There is a failure to develop optimal peak bone mass and low Vitamin D levels contribute to the development of osteopenia without associated bone loss.
2. **Secondary osteoporosis**- Steroid overuse, lactose intolerance, alcohol abuse, Renal failure, liver failure, cushing's syndrome, hyperthyroidism, hyperparathyroidism, use of anticonvulsant, antiestrogens, androgen inhibitors and proton pump inhibitors

Assessment Findings

1. Low stature with kyphosis
2. Fracture
   a. Femur
3. Bone pain

LABORATORY FINDINGS

1. **DEXA-scan (Dual Energy Xray Absorptiometry)**
   • Provides information about bone mineral density
   • *T-score is at least 2.5 Standard Deviation below the young adult mean value*

2. X-ray studies

Medical management

1. Diet therapy with calcium, flouride and Vitamin D
2. Hormone replacement therapy estrogen
3. Biphosphonates- (Alendronate, risedronate) increased bone mass by inhibiting the OSTEOCLAST
4. Moderate weight bearing exercises
5. Management of fractures
6. Calcitonin- slow down bone density loss and increase bone mass

Nursing Interventions

- Promote understanding of osteoporosis and the treatment regimen
- Provide adequate dietary supplement of calcium (1 gram /day for men 50-70 years old) 1.2 gram /day for women and vitamin D
- Instruct to employ a regular program of moderate exercises and physical activity
- Manage the constipating side-effect of calcium supplements
- Take calcium supplements with meals
- Take alendronate with an EMPTY stomach with water

- Instruct on intake of Hormonal replacement

Relieve the pain

- Instruct the patient to rest on a firm mattress
- Suggest that knee flexion will cause relaxation of back muscles
- Heat application may provide comfort
- Encourage good posture and body mechanics
- Instruct to avoid twisting and heavy lifting

Improve bowel elimination

- Constipation is a problem of calcium supplements and immobility
- Advise intake of HIGH fiber diet and increased fluids

Prevent injury

- Instruct to use isometric exercise to strengthen the trunk muscles
- AVOID sudden jarring, bending and strenuous lifting
- Provide a safe environment